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(54) **WORKING MACHINE AND METHOD OF CONTROLLING WORKING MACHINE**

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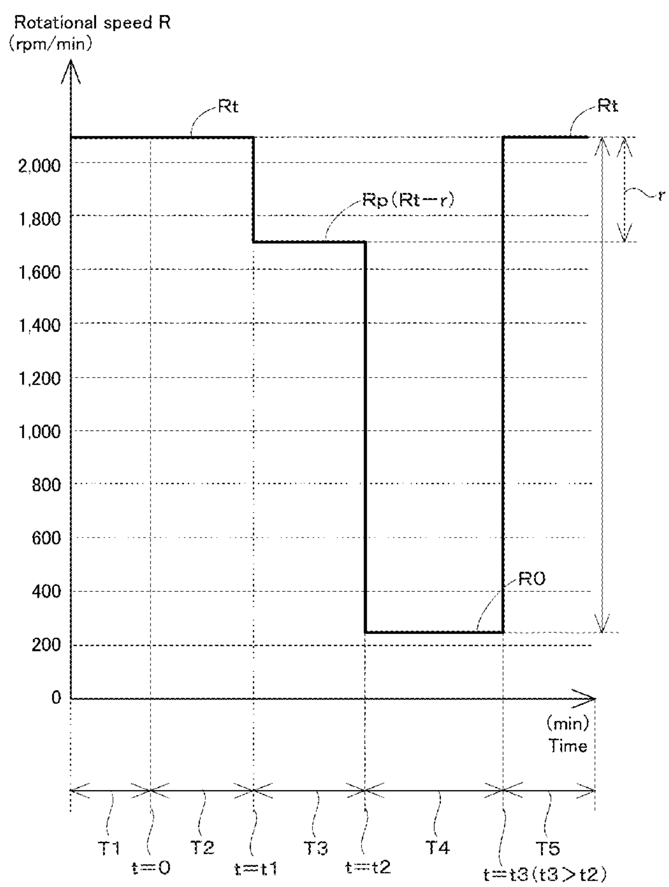
(57) **ABSTRACT**

A working machine includes a machine body, a rotary drive source, a hydraulic device to be actuated by power generated by the rotary drive source, a working device to be actuated by a hydraulic of hydraulic fluid from the hydraulic device, a work detector to detect whether the working device is actuated, and a controller configured or programmed to, when an elapsed time period which starts when the working device enters a non-actuated state reaches a first threshold, reduce a rotational speed of the rotary drive source to a reduced rotational speed obtained by subtracting a predetermined value from a set rotational speed at a time immediately before the elapsed time period reaches the first threshold or from an actual rotational speed, and, as the elapsed time period further increases from the first threshold, reduce the rotational speed to an idling rotational speed in a stepwise or continuous manner.



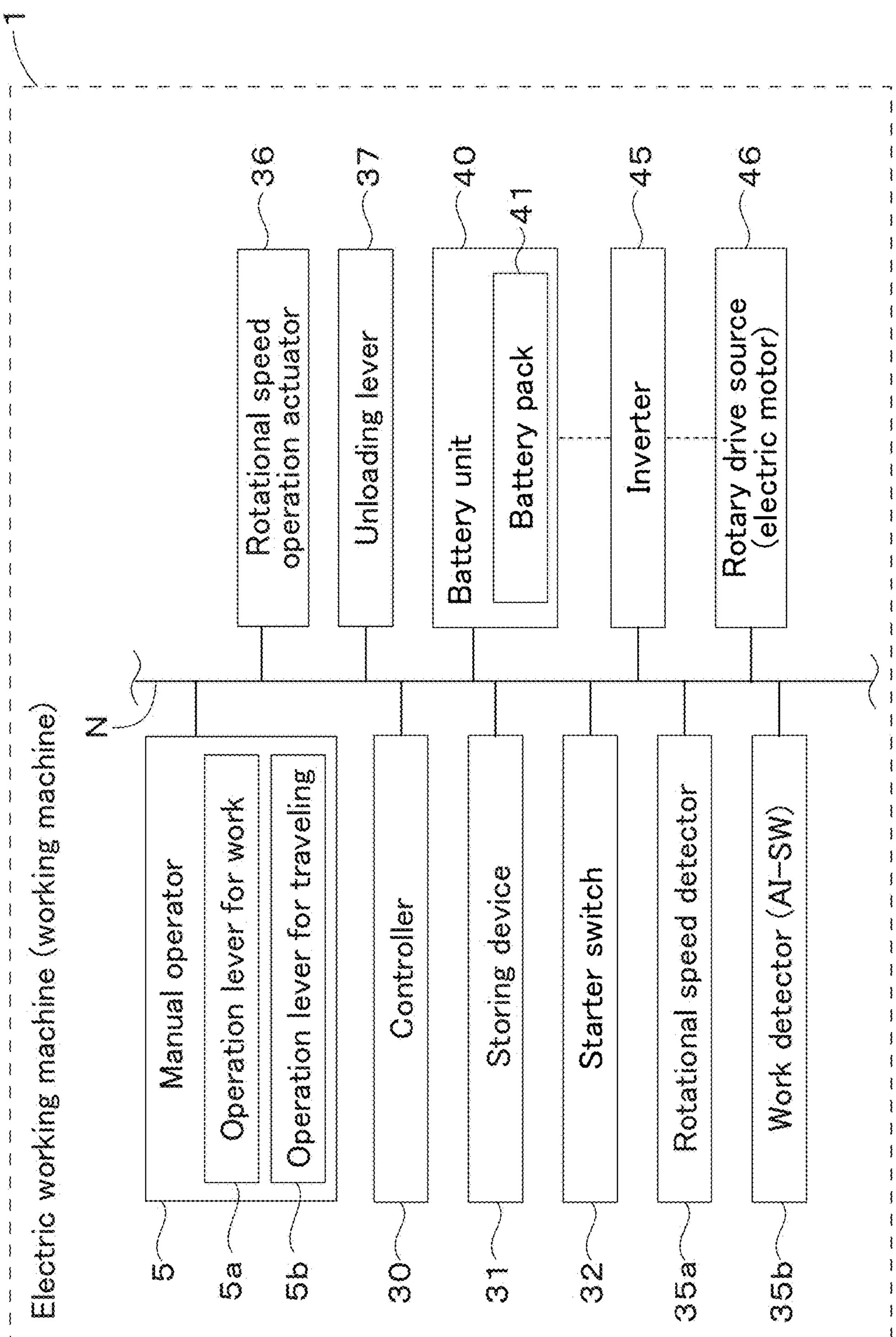

1
Electric working machine (working machine)
N
5
Manual operator
5a
Operation lever for work
5b
Operation lever for traveling
30
Controller
31
Storing device
32
Starter switch
35a
Rotational speed detector
35b
Work detector (AI-SW)
36
Rotational speed operation actuator
37
Unloading lever
40
Battery unit
41
Battery pack
45
Inverter
46
Rotary drive source (electric motor)

Fig.1

Fig.2

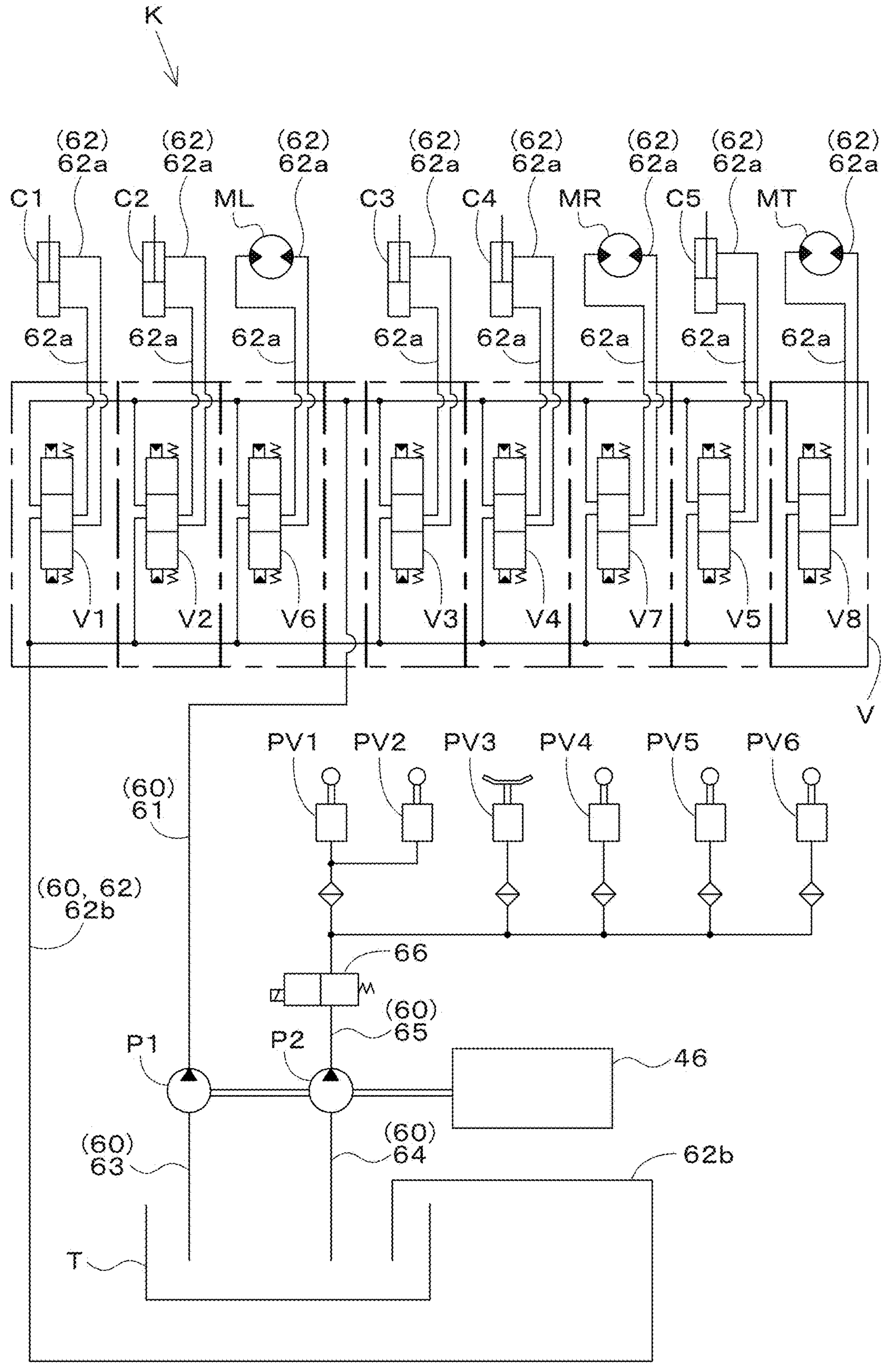

K
(62)
62a
C1
C2
ML
C3
C4
MR
C5
MT
V1
V2
V6
V3
V4
V7
V5
V8
V
PV1
PV2
PV3
PV4
PV5
PV6
(60)
61
(60, 62)
62b
66
(60)
65
P1
P2
46
(60)
63
(60)
64
62b
T

Fig.3

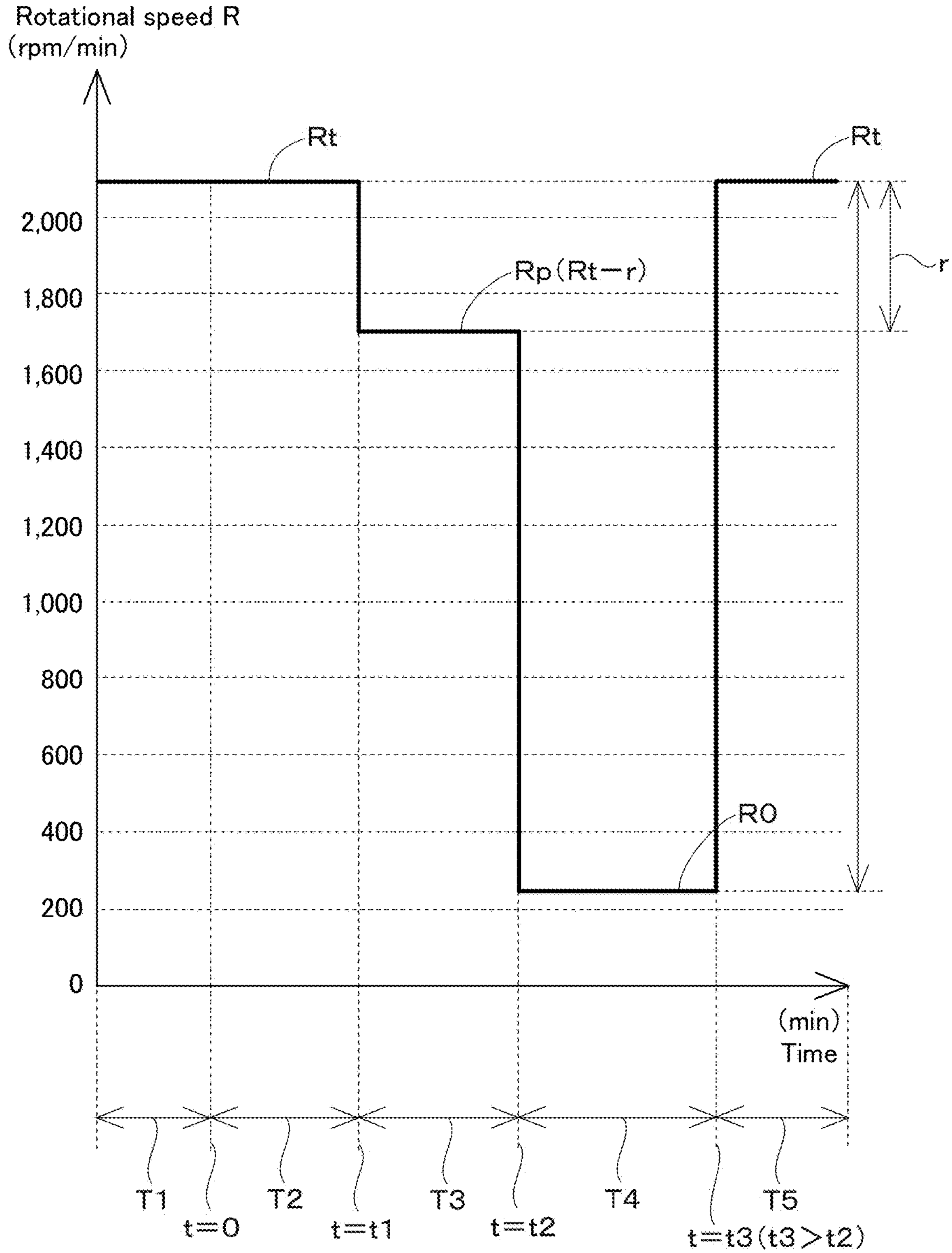
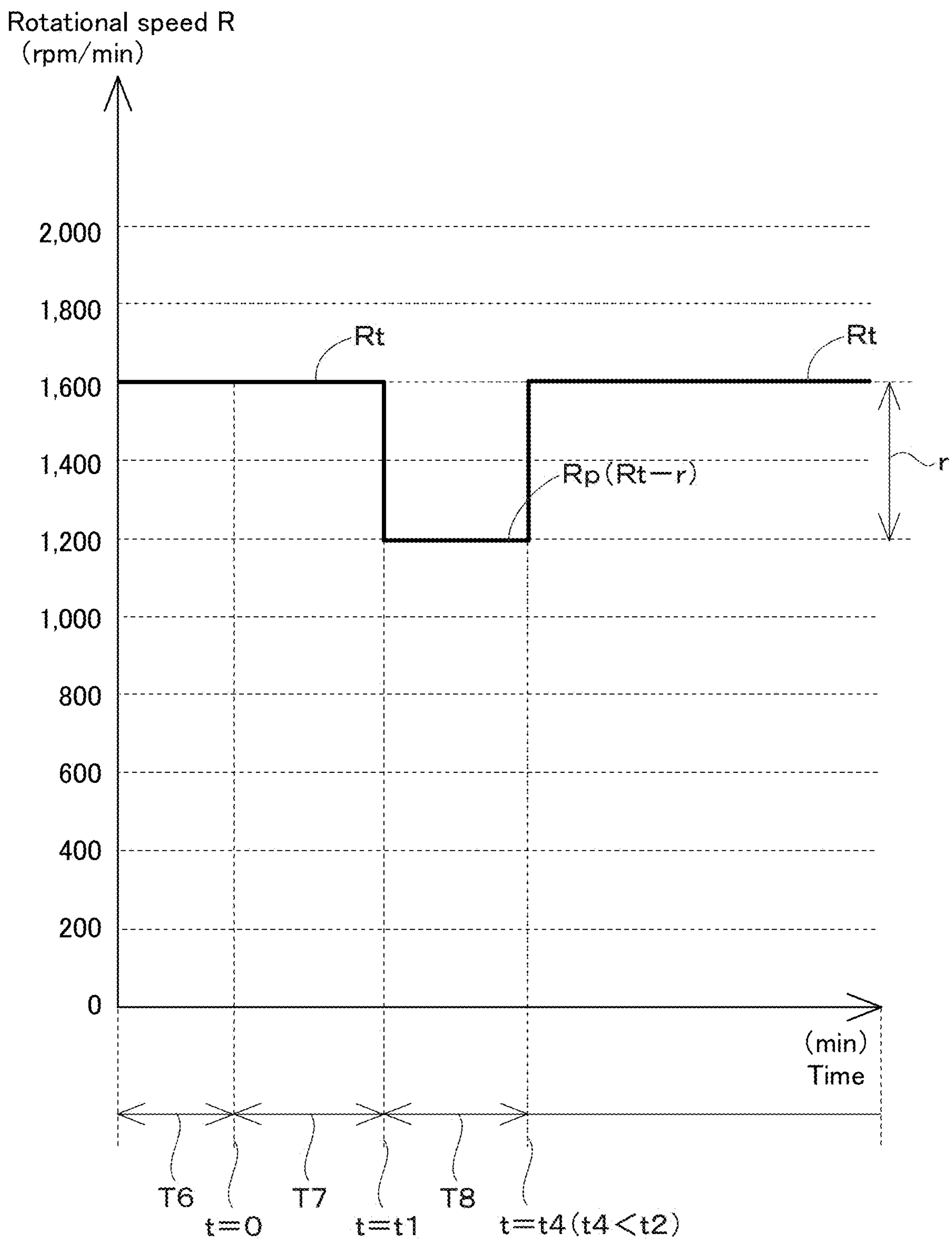

Rotational speed R
(rpm/min)
Rt
Rt
Rp(Rt−r)
r
R0
2,000
1,800
1,600
1,400
1,200
1,000
800
600
400
200
0
(min)
Time
T1
t=0
T2
t=t1
T3
t=t2
T4
t=t3(t3>t2)
T5

Fig.4


Rotational speed R
(rpm/min)
2,000
1,800
1,600
1,400
1,200
1,000
800
600
400
200
0
Rt
Rt
Rp(Rt−r)
r
(min)
Time
T6
t=0
T7
t=t1
T8
t=t4(t4<t2)

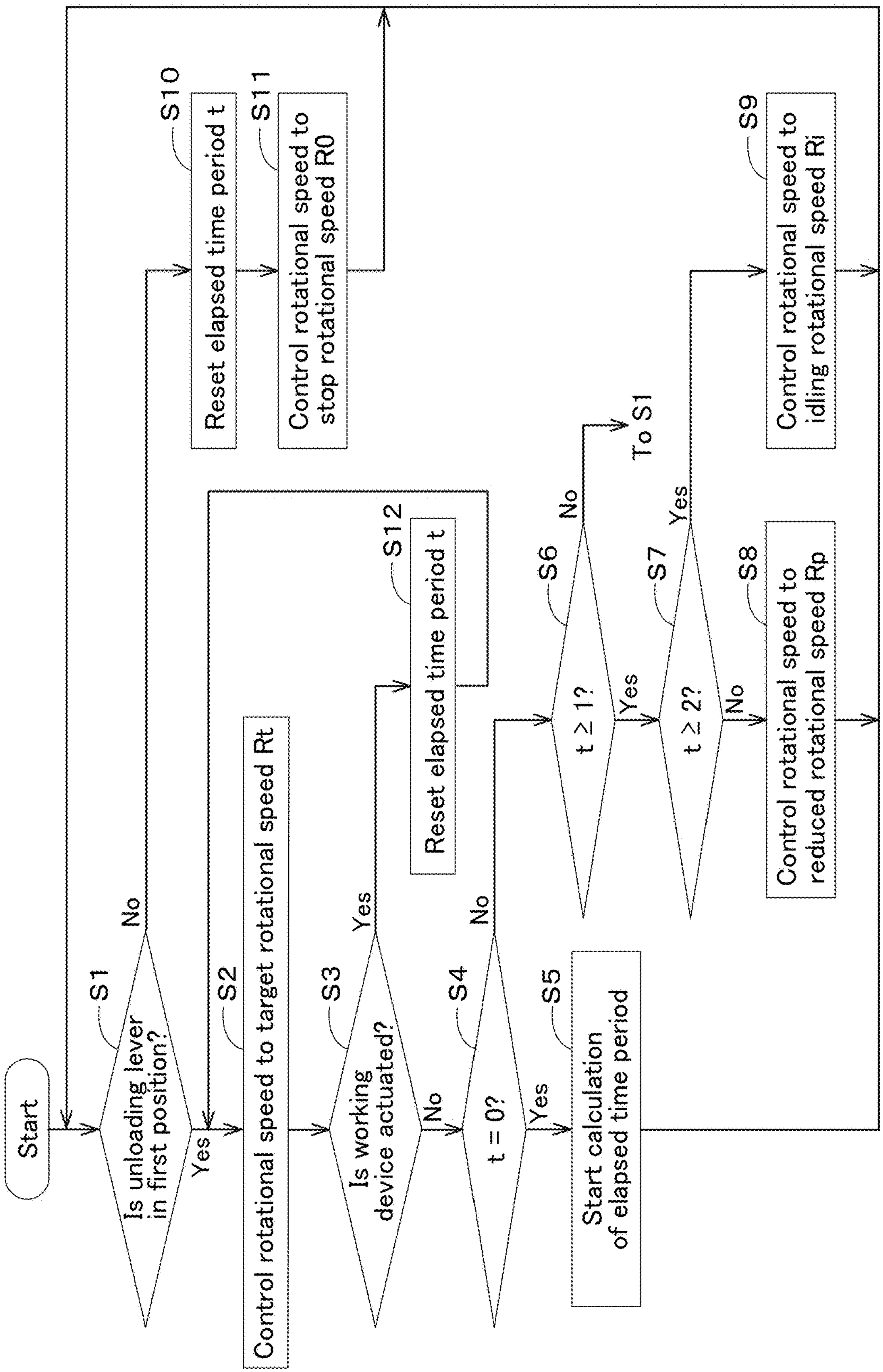

Start
S1
Is unloading lever in first position?
Yes
No
S2
Control rotational speed to target rotational speed Rt
S3
Is working device actuated?
No
Yes
S4
t = 0?
No
Yes
S5
Start calculation of elapsed time period
S6
t ≥ 1?
Yes
No
To S1
S7
t ≥ 2?
Yes
No
S8
Control rotational speed to reduced rotational speed Rp
S9
Control rotational speed to idling rotational speed Ri
S10
Reset elapsed time period t
S11
Control rotational speed to stop rotational speed R0
S12
Reset elapsed time period t

Fig.5

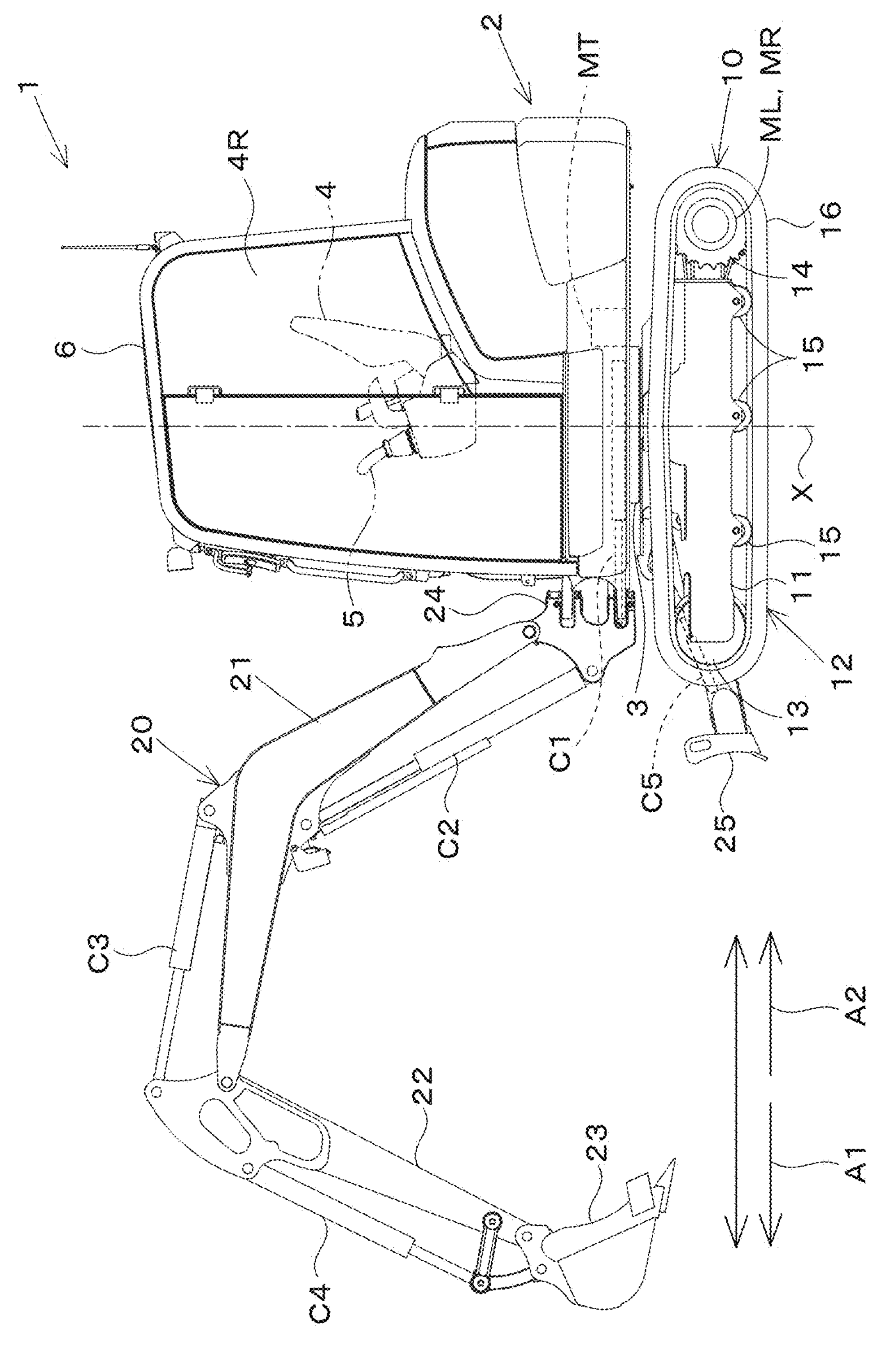

1
2
MT
10
ML, MR
4R
4
6
16
14
15
X
15
5
24
11
12
21
20
3
13
C1
C5
25
C2
C3
A2
22
23
A1
C4

Fig.6

WORKING MACHINE AND METHOD OF CONTROLLING WORKING MACHINE

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation application of International Application No. PCT/JP2023/040966, filed on Nov. 14, 2023, which claims the benefit of priority to Japanese Patent Application No. 2022-211307, filed on Dec. 28, 2022. The entire contents of each of these applications are hereby incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present invention relates to working machines and methods of controlling working machines.

2. Description of the Related Art

[0003] For example, Japanese Unexamined Patent Application Publication No. 2021-80707 discloses an electric working machine that is driven by power of an electric motor. The electric working machine (working machine) disclosed in Japanese Unexamined Patent Application Publication No. 2021-80707 includes an electric motor that is driven by electric power that is output by a battery unit, a hydraulic pump that delivers a hydraulic fluid by being driven by the electric motor, a hydraulic device that is driven by the hydraulic fluid delivered by the hydraulic pump, a working device that is actuated by the hydraulic device, a manual operator to operate the hydraulic device, a controller that controls a rotational speed of the electric motor, and the like. The controller sets the rotational speed of the electric motor in accordance with an operation of the manual operator in a case where a value of an output current from the battery unit is equal to or greater than a predetermined value and sets the rotational speed of the electric motor to a predetermined idling rotational speed in a case where the value of the output current from the battery unit is less than the predetermined value.

SUMMARY OF THE INVENTION

[0004] In a case where an electric motor is being driven to rotate not only in a case where the working device is performing work, but also in a case where a working device is not performing work, electric power of a battery unit is wasted, and work efficiency of a working machine decreases.

[0005] Example embodiments of the present invention make it possible to save energy of working machines.

[0006] A working machine according to an example embodiment of the present invention includes a machine body, a rotary drive source, a hydraulic device to be actuated by power generated by the rotary drive source, a working device to be actuated by a hydraulic pressure of hydraulic fluid supplied from the hydraulic device, a work detector to detect whether or not the working device is actuated, and a controller configured or programmed to, when an elapsed time period which starts when the working device enters a non-actuated state reaches a first threshold, reduce a rotational speed of the rotary drive source to a reduced rotational speed obtained by subtracting a predetermined value from a set rotational speed at a time immediately before the elapsed time period reaches the first threshold or from an actual rotational speed, and, as the elapsed time period exceeds the first threshold and further increases, reduce the rotational speed of the rotary drive source from the reduced rotational speed to an idling rotational speed in accordance with the elapsed time period in a stepwise or continuous manner.

[0007] The working machine may further include a manual operator to be operated to control the working device. The work detector may be operable to detect whether or not the working device is actuated based on the operation of the manual operator.

[0008] The working machine may further include a rotational speed operation actuator to be operated to set the set rotational speed. The controller may be configured or programmed to control the rotational speed of the rotary drive source at the set rotational speed set by operating the rotational speed operation actuator in a case that the elapsed time period is less than the first threshold, and reduce the rotational speed of the rotary drive source from the set rotational speed to the reduced rotational speed when the elapsed time period reaches the first threshold.

[0009] The working machine may further include a rotational speed detector to detect the rotational speed of the rotary drive source. The controller may be configured or programmed to, when the elapsed time period reaches the first threshold, reduce the rotational speed of the rotary drive source to the reduced rotational speed from a rotational speed of the rotary drive source detected by the rotational speed detector immediately before the elapsed time period reaches the first threshold.

[0010] The controller may be configured or programmed to, after the elapsed time period reaches the first threshold and the rotational speed of the rotary drive source is reduced to the reduced rotational speed, maintain the rotational speed of the rotary drive source at the reduced rotational speed until the elapsed time period reaches a second threshold greater than the first threshold, and, when the elapsed time period reaches the second threshold, further reduce the rotational speed of the rotary drive source from the reduced rotational speed.

[0011] The controller may be configured or programmed to reduce the rotational speed of the rotary drive source to the idling rotational speed in a stepwise or continuous manner after the elapsed time period reaches the second threshold.

[0012] The idling rotational speed may be a rotational speed of the rotary drive source for a case where the working device does not perform work, and the rotary drive source driven at the idling rotational speed may not generate enough power to cause the hydraulic device to actuate the working device.

[0013] The controller may be configured or programmed to, when the controller determines that the manual operator is operated when the elapsed time period is equal to or greater than the first threshold, control the rotational speed of the rotary drive source to the set rotational speed at a time immediately before the elapsed time period reaches the first threshold.

[0014] The rotary drive source may be an electric motor to be driven by electric power or an engine to be driven by burning fuel.

[0015] A method of controlling a working machine according to an example embodiment of the present invention is a method of controlling a working machine including a machine body, a rotary drive source, a hydraulic device to be actuated by power generated by the rotary drive source, and a working device to be actuated by a hydraulic pressure of hydraulic fluid supplied from the hydraulic device, the method including a first step including, when an elapsed time period which starts when the working device enters a non-actuated state reaches a first threshold, reducing a rotational speed of the rotary drive source to a reduced rotational speed obtained by subtracting a predetermined value from a set rotational speed at a time immediately before the elapsed time period reaches the first threshold or from an actual rotational speed, and a second step including, as the elapsed time period exceeds the first threshold and further increases, reducing the rotational speed of the rotary drive source from the reduced rotational speed to an idling rotational speed in accordance with the elapsed time period in a stepwise or continuous manner.

[0016] The above and other elements, features, steps, characteristics and advantages of the present invention will become more apparent from the following detailed description of the example embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] A more complete appreciation of example embodiments of the present invention and many of the attendant advantages thereof will be readily obtained as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings described below.

[0018] FIG. **1** is an electric block diagram of a working machine.

[0019] FIG. **2** is a hydraulic circuit diagram of the working machine.

[0020] FIG. **3** illustrates a first example of a rotational speed of an electric motor in a case where auto idling control is performed.

[0021] FIG. **4** illustrates a second example of a rotational speed of an electric motor in a case where auto idling control is performed.

[0022] FIG. **5** is a view for explaining a flow of a series of processes of control of a rotational speed of the electric motor performed by a controller.

[0023] FIG. **6** is an overall side view of the working machine.

DETAILED DESCRIPTION OF THE EXAMPLE EMBODIMENTS

[0024] Example embodiments will now be described with reference to the accompanying drawings, wherein like reference numerals designate corresponding or identical elements throughout the various drawings. The drawings are to be viewed in an orientation in which the reference numerals are viewed correctly.

[0025] Example embodiments of the present invention are described below with reference to the drawings.

[0026] First, an overall configuration of a working machine **1** is described.

[0027] FIG. **6** is an overall side view of the working machine **1**. The working machine **1** is a backhoe. In the present example embodiment, the working machine **1** is described by taking, as an example, an electric working machine that operates by electric power supplied from a battery unit **40**. The working machine **1** includes a machine body (swivel base) **2**, a traveling device **10**, a working device **20**, and the like. On the machine body **2**, an operator's seat **4** on which a user sits and a protection mechanism **6** that protects the operator's seat **4** from front, rear, left, right and upper sides are provided. A manual operator **5** to operate the working machine **1** is provided around the operator's seat **4**.

[0028] In the following description, a direction (a direction indicated by arrow A**1** in FIG. **6**) which a user sitting on the operator's seat **4** faces is referred to as a forward direction, and an opposite direction (a direction indicated by arrow A**2** in FIG. **6**) is referred to as a rearward direction. Furthermore, a left side viewed from a user's point of view (the near side in FIG. **6**) is referred to as a leftward direction, and a right side viewed from the user's point of view (the far side in FIG. **6**) is referred to as a rightward direction. A horizontal direction orthogonal to a front-rear direction (machine body front-rear direction) is referred to as a machine body width direction.

[0029] The traveling device **10** is a device that allows the machine body **2** to travel, and includes a traveling frame (truck frame) **11** and a traveling mechanism **12**. The traveling frame **11** is a structure around which the traveling mechanism **12** is attached and on which the machine body **2** is supported.

[0030] The traveling mechanism **12** is, for example, a crawler-type traveling mechanism. The traveling mechanism **12** is provided on the left and right of the traveling frame **11**. The traveling mechanism **12** includes an idler **13**, a drive wheel **14**, a plurality of track rollers **15**, an endless crawler belt **16**, and a traveling motor ML or MR.

[0031] The idler **13** is provided on a front portion of the traveling frame **11**. The drive wheel **14** is provided on a rear portion of the traveling frame **11**. The plurality of track rollers **15** are provided between the idler **13** and the drive wheel **14**. The crawler belt **16** is suspended over the idler **13**, the drive wheel **14**, and the track rollers **15**.

[0032] The left traveling motor ML is included in the traveling mechanism **12** on the left of the traveling frame **11**. The right traveling motor MR is included in the traveling mechanism **12** on the right of the traveling frame **11**. These traveling motors ML and MR are hydraulic motors. In each traveling mechanism **12**, the drive wheel **14** is driven by power of the traveling motor ML or MR to rotate, and the crawler belt **16** thus circulates in a circumferential direction.

[0033] The machine body **2** is supported on the traveling frame **11** with a swivel bearing **3** interposed therebetween so as to be rotatable about a swivel axis X. A swivel motor MT is provided in the machine body **2**. The swivel motor MT is a hydraulic motor (hydraulic actuator). The machine body **2** swivels about the swivel axis X by power of the swivel motor MT.

[0034] The working device **20** is supported on a front portion of the machine body **2**. The working device **20** includes a boom **21**, an arm **22**, a bucket (working tool) **23**, a dozer device **25**, and hydraulic cylinders (hydraulic actuators) C1 to C5. A base end of the boom **21** is pivotally attached to a swing bracket **24** so as to be rotatable about a horizontal axis (an axis extending in the machine body width direction). Accordingly, the boom **21** is swingable in an up-down direction (vertical direction). The arm **22** is pivotally attached to a leading end of the boom **21** so as to be rotatable about a horizontal axis. Accordingly, the arm **22** is swingable in the front-rear direction or the up-down direction. The bucket **23** is provided on a leading end of the arm **22** so as to be capable of performing a shoveling action and a dumping action.

[0035] To the working machine **1**, another working tool (hydraulic attachment) that can be driven by a hydraulic actuator can be attached instead of or in addition to the bucket **23**. Examples of such a working tool include a hydraulic breaker, a hydraulic crusher, an angle broom, an earth auger, a pallet fork, a sweeper, a mower, and a snow blower.

[0036] The swing bracket **24** swings leftward and rightward by extension and contraction of the swing cylinder C**1** provided in the machine body **2**. The boom **21** swings upward and downward (forward and rearward) by extension and contraction of the boom cylinder C**2**. The arm **22** swings upward and downward (forward and rearward) by extension and contraction of the arm cylinder C**3**. The bucket **23** performs a shoveling action and a dumping action by extension and contraction of the bucket cylinder (working tool cylinder) C**4**. The swing cylinder C**1**, the boom cylinder C**2**, the arm cylinder C**3**, and the bucket cylinder C**4** are hydraulic cylinders (hydraulic actuators).

[0037] The dozer device **25** is attached to a front portion of the traveling device **10**. The dozer device **25** swings upward and downward by extension and contraction of the dozer cylinder C**5**. The dozer cylinder C**5** is attached to the traveling frame **11**. The dozer cylinder C**5** is a hydraulic cylinder (hydraulic actuator).

[0038] The working machine **1** performs work while traveling by the traveling device **10** including the traveling motors ML and MR and swiveling the machine body **2** by the working device **20** including the hydraulic cylinders C**1** to C**5** and the swivel motor MT. That is, it can be said that not only the working device **20**, but also the traveling device **10** and the swivel motor MT are working devices **20** included in the working machine **1**. Hereinafter, for convenience of description, the working device **20** and the traveling device **10** are sometimes collectively referred to as "working devices **20** and **10**". Note that hydraulic actuators such as the traveling motors ML and MR, the swivel motor MT, and the hydraulic cylinders C**1** to C**5** are included in hydraulic devices.

[0039] Next, an electric configuration of the working machine **1** is described. FIG. **1** is an electric block diagram of the working machine **1** according to the first example embodiment. As illustrated in FIG. **1**, the working machine **1** includes a controller **30**, a storing device (memory and/or storage) **31**, a battery unit **40**, an inverter **45**, and a rotary drive source **46**.

[0040] The controller **30** is provided in the machine body **2** or inside the protection mechanism **6** and is a device including an electric/electronic circuit, a CPU, a program stored in a memory, and the like. The controller **30** controls various devices connected to an in-vehicle network N of the working machine **1**. The controller **30** control operation of units included in the working machine **1** such as the units illustrated in FIG. **1**. For example, the controller **30** can control drive of the rotary drive source **46**.

[0041] A starter switch **32** is connected to the controller **30**. The starter switch **32** is provided inside the protection mechanism **6**, and the user sitting on the operator's seat **4** can operate the starter switch **32**. The starter switch **32** is operated to start or stop the working machine **1**. The controller **30** starts each unit included in the working machine **1** in response to an ON operation of the starter switch **32** and stops each unit included in the working machine **1** in response to an OFF operation of the starter switch **32**.

[0042] The storing device **31** is a recording medium such as a solid state drive (SSD) or a hard disk drive (HDD) and stores therein various kinds of information concerning the working machine **1**.

[0043] The battery unit **40** is a structure that can store electric power, and discharges (outputs) the stored electric power. The battery unit **40** is mounted on the machine body **2**. The battery unit **40** includes a battery pack **41**. The battery pack **41** is a secondary battery (rechargeable battery) such as a lithium-ion battery including at least one battery.

[0044] The inverter **45** is electrically connected to the battery unit **40** and the electric motor **46**, and converts DC electric power input from the battery unit **40** into three-phase AC electric power and supplies the three-phase AC electric power to the electric motor **46**. Furthermore, the inverter **45** can freely adjust a frequency and a voltage of electric power supplied to the electric motor **46**.

[0045] In the present example embodiment, the rotary drive source **46** is an electric motor that generates power by being driven by supplied electric power. The electric motor **46** receives electric power from the battery unit **40** (the battery pack **41**). The electric motor **46** is electrically connected to the inverter **45** and is driven by electric power supplied from the battery unit **40** via the inverter **45**. The electric motor **46** is a drive source of the working machine **1** and is, for example, a permanent magnet embedded three-phase AC synchronous motor. The electric motor **46** includes a rotor that is rotatable and a stator that generates force for rotating the rotor.

[0046] Note that the electric motor **46** may be another kind of synchronous motor, and may be AC motor or may be a DC motor.

[0047] The manual operator **5** includes an operation member for operating the working device **20**. Specifically, the manual operator **5** includes operation members such as an operation lever **5***a* for work and an operation lever **5***b* for traveling. The manual operator **5** may also include a potentiometer, a switch, a sensor, or the like (not illustrated) for detecting whether or not the operation levers **5***a* and **5***b* have been operated, operation positions of the operation levers **5***a* and **5***b*, or operation amounts of the operation levers **5***a* and **5***b*.

[0048] The operation lever **5***a* for work is a member for operating actuation of the working device **20**. The operation lever **5***b* for traveling is a member for operating actuation of the traveling device **10**. Although the operation lever **5***a* for work and the operation lever **5***b* for traveling are each illustrated as a single block for convenience of description in FIG. **1**, a plurality of operation levers **5***a* for work and a plurality of operation levers **5***b* for traveling are actually provided.

[0049] Next, a hydraulic circuit K included in the working machine **1** is described. FIG. **2** is a hydraulic circuit diagram of the working machine **1**. As illustrated in FIG. **2**, hydraulic devices such as the hydraulic actuators C**1** to C**5**, ML, MR, and MT, a control valve V, hydraulic pumps P**1** and P**2**, a hydraulic fluid tank T, operation valves PV**1** to PV**6**, an unloading valve **66**, and a fluid passage **60** are provided in the hydraulic circuit K. The hydraulic devices provided in the hydraulic circuit K of the working machine **1** are actuated by power generated by the electric motor **46**. Specifically, one of the hydraulic pumps P**1** and P**2** is a hydraulic pump P**1** for actuation, and the other one of the hydraulic pumps P**1** and P**2** is a hydraulic pump P**2** for control. These hydraulic pumps P**1** and P**2** are driven by power of the electric motor **46**.

[0050] The hydraulic pump P**1** for actuation sucks a hydraulic fluid stored in the hydraulic fluid tank T and delivers the hydraulic fluid to the control valve V. Although a single hydraulic pump P**1** for actuation is illustrated in FIG. **2** for convenience of description, this is not restrictive, and an appropriate number of hydraulic pumps P**1** for actuation that supply the hydraulic fluid to the hydraulic actuators C**1** to C**5**, ML, MR, and MT may be provided.

[0051] The hydraulic pump P**2** for control outputs a hydraulic pressure for a signal, a hydraulic pressure for control, or the like by delivering the hydraulic fluid in the hydraulic fluid tank T. That is, the hydraulic pump P**2** for control supplies (delivers) a pilot oil. An appropriate number of hydraulic pumps P**2** for control may be provided.

[0052] The control valve V includes a plurality of controlling valves V**1** to V**8**. The controlling valves V**1** to V**8** control (adjust) a flow rate of the hydraulic fluid output from the hydraulic pumps P**1** and P**2** to the hydraulic actuators C**1** to C**5**, ML, MR, and MT. The swing controlling valve V**1** controls a flow rate of a hydraulic fluid supplied to the swing cylinder C**1**. The boom controlling valve V**2** controls a flow rate of a hydraulic fluid supplied to the boom cylinder C**2**. The arm controlling valve V**3** controls a flow rate of a hydraulic fluid supplied to the arm cylinder C**3**. The bucket controlling valve V**4** controls a flow rate of a hydraulic fluid supplied to the bucket cylinder C**4**. The dozer controlling valve V**5** controls a flow rate of a hydraulic fluid supplied to the dozer cylinder C**5**. The traveling controlling valve V**6** for left controls a flow rate of a hydraulic fluid supplied to the traveling motor ML on the left. The traveling controlling valve V**7** for right controls a flow rate of a hydraulic fluid supplied to the traveling motor MR on the right. The swivel controlling valve V**8** controls a flow rate of a hydraulic fluid supplied to the swivel motor MT.

[0053] The operation valves (remote control valves) PV**1** to PV**6** operate in accordance with an operation of various operation levers (the operation lever **5***a* for work and the operation lever **5***b* for traveling) included in the manual operator **5**. A pilot oil acts on the controlling valves V**1** to V**8** in proportion to actuation amounts (operation amounts) of the operation valves PV**1** to PV**6**, and thereby spools of the controlling valves V**1** to V**8** are moved. A hydraulic fluid of amounts proportional to amounts by which the spools of the controlling valves V**1** to V**8** are moved is supplied to the hydraulic actuators C**1** to C**5**, ML, MR, and MT to be controlled. Furthermore, the hydraulic actuators C**1** to C**5**, ML, MR, and MT are driven in accordance with amounts of hydraulic fluid supplied from the controlling valves V**1** to V**8**.

[0054] In other words, by operating the operation levers **5***a* and **5***b*, the hydraulic fluid (pilot oil) that acts on the controlling valves V**1** to V**8** is adjusted, and thereby the controlling valves V**1** to V**8** are controlled. This adjusts amounts of the hydraulic fluid supplied from the controlling valves V**1** to V**8** to the hydraulic actuators C**1** to C**5**, ML, MR, and MT, thereby controlling drive and stoppage of the hydraulic actuators C**1** to C**5**, ML, MR, and MT. In this way, the working devices **20** and **10** are actuated by a hydraulic pressure of the hydraulic fluid supplied from a hydraulic device (the hydraulic pump P**1**).

[0055] The fluid passage **60** is a flow passage that connects units provided in the hydraulic circuit K and through which the hydraulic fluid or pilot oil flows to the units. The fluid passage **60** includes a first fluid passage **61**, a second fluid passage **62**, a first suction fluid passage **63**, a second suction fluid passage **64**, and a restriction fluid passage **65**.

[0056] The first suction fluid passage **63** is a flow passage through which the hydraulic fluid sucked from the hydraulic fluid tank T by the hydraulic pump P**1** for actuation flows. The second suction fluid passage **64** is a flow passage through which the hydraulic fluid sucked from the hydraulic fluid tank T by the hydraulic pump P**2** for control flows.

[0057] The first fluid passage **61** is a flow passage through which the hydraulic fluid delivered by the hydraulic pump P**1** for actuation flows toward the controlling valves V**1** to V**8** of the control valve V. The first fluid passage **61** branch into a plurality of fluid passages in the control valve V, which are connected to the controlling valves V**1** to V**8**.

[0058] The second fluid passage **62** is a flow passage through which the hydraulic fluid that has passed the controlling valves V**1** to V**8** flows toward the hydraulic fluid tank T. The hydraulic fluid tank T stores the hydraulic fluid. The second fluid passage **62** includes a reciprocating fluid passage **62***a* and a discharge fluid passage **62***b*.

[0059] A plurality of reciprocating fluid passages **62***a* are provided so that each of the controlling valves V**1** to V**8** and a corresponding one of the hydraulic actuators C**1** to C**5**, ML, MR, and MT to be controlled are connected by a pair of two reciprocating fluid passages **62***a*. The reciprocating fluid passages **62***a* are flow passages through which the hydraulic fluid is supplied from the controlling valves V**1** to V**8** to the hydraulic actuators C**1** to C**5**, ML, MR, and MT and the hydraulic fluid is returned from the hydraulic actuators C**1** to C**5**, ML, MR, and MT to the controlling valves V**1** to V**8**. One end of the discharge fluid passage **62***b* branch into a plurality of discharge fluid passages, which are connected to the controlling valves V**1** to V**8**. The other end portion of the discharge fluid passage **62***b* is connected to the hydraulic fluid tank T.

[0060] A part of the hydraulic fluid that has flown to any one of the controlling valves V**1** to V**8** through the first fluid passage **61** passes the one of the controlling valves V**1** to V**8**, passes through one of the reciprocating fluid passages **62***a*, and is supplied to a corresponding one of the hydraulic actuators C**1** to C**5**, ML, MR, and MT to be controlled. Then, the hydraulic fluid discharged from the corresponding one of the hydraulic actuators C**1** to C**5**, ML, MR, and MT returns to the connected one of the controlling valves V**1** to V**8** by passing through the other one of the reciprocating fluid passages **62***a*, passes the one of the controlling valves V**1** to V**8**, and flows to the discharge fluid passage **62***b*.

[0061] The other portion of the hydraulic fluid that has flown to any one of the controlling valves V**1** to V**8** by passing through the first fluid passage **61** passes the one of the controlling valves V**1** to V**8**, flows to the discharge fluid passage **62***b*, and returns to the hydraulic fluid tank T without being supplied to the hydraulic actuators C**1** to C**5**, ML, MR, and MT. As described above, the fluid passages **61**, **62**, and **63** are provided so that the hydraulic fluid circulates through the hydraulic fluid tank T, the hydraulic pump P**1**, and the controlling valves V**1** to V**8** of the control valve V, (a part of the hydraulic fluid also circulates through the hydraulic actuators C**1** to C**5**, ML, MR, and MT).

[0062] The restriction fluid passage **65** is a flow passage through which the hydraulic fluid delivered by the hydraulic pump P**2** for control flows to the operation valves PV**1** to PV**6**. One end portion of the restriction fluid passage **65** is connected to the hydraulic pump P**2** for control, and the other end of the restriction fluid passage **65** branches into a plurality of fluid passages, which are connected to primary-side ports (primary ports) of the operation valves PV**1** to PV**6**.

[0063] The unloading valve **66** is provided on the restriction fluid passage **65**. The unloading valve **66** prohibits drive of the hydraulic actuators C**1** to C**5**, ML, MR, and MT, that is, drive of the working devices **20** and **10** by cutting off supply of the hydraulic fluid from the hydraulic pump P**1** for actuation to the hydraulic actuators C**1** to C**5**, ML, MR, and MT.

[0064] The unloading valve **66** is switched between a supply position and a cut-off position in accordance with an operation of an unloading lever **37**. The controller **30** outputs an instruction signal to the unloading valve **66** based on an operation of the unloading lever **37**, and the unloading valve **66** is thus switched between the supply position and the cut-off position. The unloading valve **66** is biased toward the cut-off position (unloading position) by a spring, and the unloading valve **66** is in the cut-off position when a solenoid is deenergized, and the unloading valve **66** is switched to the supply position when the solenoid is energized. The unloading valve **66** is energized when the unloading lever **37** is lowered, and the unloading valve **66** is deenergized by raising the unloading lever **37**.

[0065] When the unloading valve **66** is switched to the supply position, the hydraulic fluid delivered from the hydraulic pump P**2** for control to the restriction fluid passage **65** is supplied to the operation valves PV**1** to PV**6**, which allows the controlling valves V**1** to V**8** to be operated. This also allows the hydraulic actuators C**1** to C**5**, ML, MR, and MT, and the working devices **20** and **10** to be operated. The hydraulic fluid discharged from the operation valves PV**1** to PV**6** returns to the hydraulic fluid tank T by passing through another discharge fluid passage (not illustrated).

[0066] On the other hand, when the unloading valve **66** is switched to the cut-off position, the hydraulic fluid delivered from the hydraulic pump P**2** for control to the restriction fluid passage **65** is discharged to the hydraulic fluid tank T and ceases to be supplied to the operation valves PV**1** to PV**6** (supply of the hydraulic fluid is stopped), which prohibits the controlling valves V**1** to V**8** from being operated. This also prohibits the hydraulic actuators C**1** to C**5**, ML, MR, and MT, and the working devices **20** and **10** from being operated.

[0067] Although a case where the operation valves PV**1** to PV**6** and the controlling valves V**1** to V**8** are provided in the hydraulic circuit K and the operation valves PV**1** to PV**6** adjust a pilot oil that acts on the controlling valves V**1** to V**8** when the operation levers **5***a* and **5***b* are operated has been described as an example in the above example embodiment, the controlling valves V**1** to V**8** may adjust a hydraulic fluid that is supplied to the hydraulic actuators C**1** to C**5**, ML, MR, and MT to be controlled based on an instruction signal from the controller **30**. In this case, the controller **30** acquires an operation signal from the operation levers **5***a* and **5***b* and outputs an instruction signal to the controlling valves V**1** to V**8** based on the operation signal.

[0068] Next, control of the rotary drive source (electric motor) **46** performed by the controller **30** is described. The controller **30** controls a rotational speed (motor rotational speed) R of the electric motor **46** in accordance with states of a rotational speed operation actuator (accelerator dial) **36**, a work detector **35***b*, and the unloading lever **37** during drive of the electric motor **46**.

[0069] As illustrated in FIG. **1**, the working machine **1** includes the rotational speed operation actuator **36**. The rotational speed operation actuator **36** is an operation member for operating the motor rotational speed R. As illustrated in FIG. **1**, the rotational speed operation actuator **36** is connected to the controller **30** and outputs an operation signal to the controller **30**. The rotational speed operation actuator **36** is, for example, a dial-type switch such as a selector switch having a plurality of switch positions, and a target value (a set rotational speed, a target rotational speed) Rt of the motor rotational speed R is allocated to each of the plurality of switch positions. The rotational speed operation actuator **36** can set the set rotational speed Rt of the electric motor **46** within a predetermined range. In the present example embodiment, the rotational speed operation actuator **36** can be operated to set the set rotational speed Rt within a range of 1000 rpm/min to 2200 rpm/min.

[0070] Although the rotational speed operation actuator **36** is a selector switch in the above example, the rotational speed operation actuator **36** may be a pedal type, a lever type, or the like as long as at least the set rotational speed Rt can be operated. The range of the set rotational speed Rt of the electric motor **46** that is operated by the rotational speed operation actuator **36** is not limited to the range of 1000 rpm/min to 2200 rpm/min. A minimum value of the set rotational speed Rt that can be set by the rotational speed operation actuator **36** is preferably defined to a lower limit value of the rotational speed R of the electric motor **46** for performing work by the working devices **20** and **10**. The lower limit value is defined to a rotational speed R of the electric motor **46** in a non-load state that can instantly actuate the working devices **20** and **10** in accordance with an operation of the operation levers **5***a* and **5***b*.

[0071] As illustrated in FIG. **1**, a rotational speed detector **35***a* that detects an actual rotational speed (actual motor rotational speed) Ra of the electric motor **46** is connected to the controller **30**, and the controller **30** controls the rotational speed R of the electric motor **46** based on the actual motor rotational speed Ra detected by the rotational speed detector **35***a* and the set rotational speed Rt set by the rotational speed operation actuator **36**. The rotational speed detector **35***a* is a sensor, an encoder, a pulse generator, or the like that detects the actual motor rotational speed Ra. The controller **30** calculates the actual motor rotational speed Ra based on a detection signal input from the rotational speed detector **35***a*.

[0072] The controller **30** transmits an instruction signal to the inverter **45** based on the calculated actual motor rotational speed Ra and the operation signal output from the rotational speed operation actuator **36**. The inverter **45** changes the motor rotational speed R of the electric motor **46** by adjusting a frequency and a voltage of electric power supplied to the electric motor **46** in accordance with the instruction signal output from the controller **30**. Specifically, the controller **30** controls drive of the electric motor **46** by the inverter **45** so that the actual motor rotational speed Ra detected by the rotational speed detector **35***a* matches the set rotational speed Rt.

[0073] As illustrated in FIG. **1**, the working machine **1** includes the work detector **35***b*. The work detector **35***b* detects whether or not the working devices **20** and **10** are working. The work detector **35***b* detects whether or not the working devices **20** and **10** are working based on an operation performed on the manual operator **5**. In the following description, a state where the working devices **20** and **10** are actuated is referred to as an actuated state, and a state where the working devices **20** and **10** are not actuated is referred to as a non-actuated state.

[0074] In the present example embodiment, the work detector **35***b* is an auto idling (AI)-switch (SW) that is a pressure sensor actuated by a hydraulic pressure of the hydraulic fluid. The AI-SW **35***b* is in an ON state in a case where at least one of the working devices **20** and **10** is in the actuated state and is in an OFF state in a case where all of the working devices **20** and **10** are in a non-actuated state. That is, the AI-SW **35***b* detects whether or not the working devices **20** and **10** are working.

[0075] The AI-SW **35***b* is provided on an operation detection fluid passage (not illustrated) for detecting operation states of the controlling valves V**1** to V**8** in the hydraulic circuit K. The operation detection fluid passage is a fluid passage through which a pilot oil delivered from the hydraulic pump P**2** for control sequentially passes a plurality of switch valves for switching positions of the controlling valves V**1** to V**8** and returns to the hydraulic fluid tank **48**. The AI-SW **35***b* is connected to an upstream side of the controlling valve V**1** provided closest to the hydraulic pump P**2** for control on the operation detection fluid passage.

[0076] When any of the controlling valves V**1** to V**8** is operated from a neutral position to a switch position, a part of the operation detection fluid passage is cut off, and a pressure of the pilot oil in the operation detection fluid passage increases to some degree (a state where the pressure has risen). Accordingly, the AI-SW **35***b* is in the ON state. That is, the AI-SW **35***b* detects that at least one of the working devices **20** and **10** is in the actuated state. When all of the controlling valves V**1** to V**8** are in the neutral position, the operation detection fluid passage is open, and therefore the pressure of the pilot oil in the operation detection fluid passage does not increase to some degree (a state where the pressure has not risen). Accordingly, the AI-SW **35***b* is in the OFF state. That is, the AI-SW **35***b* detects that all of the working devices **20** and **10** are in the non-actuated state. In this way, the work detector **35***b* detects whether or not the working device **20** is actuated based on an operation of the manual operator **5**.

[0077] Note that the work detector **35***b* is not limited to the AI-SW **35***b* provided on the operation detection fluid passage as long as the work detector **35***b* can detect whether or not the working device **20** is actuated, and the work detector **35***b* may be a potentiometer, a switch, a sensor, or the like for detecting whether or not the operation levers **5***a* and **5***b* of the manual operator **5** have been operated, operation positions of the operation levers **5***a* and **5***b*, or operation amounts of the operation levers **5***a* and **5***b*.

[0078] The controller **30** acquires a detection result of the work detector **35***b* during drive of the electric motor **46**, and reduces the motor rotational speed R in accordance with an elapsed time period t which starts when the working device **20** enters the non-actuated state and reduces the motor rotational speed R to a predetermined idling rotational speed Ri (auto idling control). In a case where the AI-SW **35***b* detects an ON state, the controller **30** determines that the manual operator **5** is being operated and the working device **20** or **10** is in the actuated state. In a case where the AI-SW **35***b* detects an OFF state, the controller **30** determines that the manual operator **5** is not being operated and the working devices **20** and **10** are in the non-actuated state.

[0079] The controller **30** calculates, as the elapsed time period t, a period from a time at which it is determined that the manual operator **5** is not being operated in a state where the electric motor **46** is being driven. The controller **30** stores the calculated elapsed time period t in the memory. When it is determined that the manual operator **5** has been operated from the detection result of the work detector **35***b*, the controller **30** returns (resets) the calculated elapsed time period t to zero (t=0).

[0080] The idling rotational speed Ri is a rotational speed (motor rotational speed) R of the rotary drive source (electric motor) **46** in a state where the working device **20** is not performing work and is defined to a value less than a lower limit value Rtmin (1000 rpm/min) of the set rotational speed Rt that can be set by the rotational speed operation actuator **36**. The rotary drive source (electric motor) **46** that is driven at the idling rotational speed Ri does not generate power enough to actuate the working devices **20** and **10** by the hydraulic device P**1**. The idling rotational speed Ri is, for example, 250 rpm/min.

[0081] Note that the idling rotational speed Ri is a rotational speed R of the electric motor **46** that can generate a hydraulic pressure of the hydraulic fluid that allows the AI-SW **35***b* to detect whether or not the working devices **20** and **10** are working. More specifically, the idling rotational speed Ri is defined to a rotational speed R of the electric motor **46** for generating a hydraulic pressure equal to or greater than a minimum hydraulic pressure of the hydraulic fluid that allows the AI-SW **35***b* to be switched to the ON state in a case where at least one of the working devices **20** and **10** is actuated and allows the AI-SW **35***b* to be switched to the OFF state in a case where the working devices **20** and **10** are not working.

[0082] The auto idling control performed by the controller **30** is described in detail below. When the calculated elapsed time period t reaches a first threshold t**1** (t=t**1**), the controller **30** decreases the motor rotational speed R, and further reduces the motor rotational speed R in accordance with the elapsed time period t. That is, in a case where the elapsed time period t is less than the first threshold t**1** (t<t**1**), the controller **30** controls the motor rotational speed R to the set rotational speed Rt set by the rotational speed operation actuator **36**, and when the elapsed time period t reaches the first threshold t**1** (t=t**1**), the controller **30** reduces the motor rotational speed R from the set rotational speed Rt set by the rotational speed operation actuator **36**.

[0083] Specifically, when the elapsed time period t reaches the first threshold t**1** (t=t**1**), the controller **30** reduces the motor rotational speed R to a reduced rotational speed Rp obtained by subtracting a predetermined value r from the set rotational speed Rt set by the rotational speed operation actuator **36** (Rp=Rt−r). The controller **30** controls the motor rotational speed R to the reduced rotational speed Rp by transmitting an instruction signal to the inverter **45** based on the calculated actual motor rotational speed Ra, an operation signal output from the rotational speed operation actuator **36**, and the predetermined value r.

[0084] In a case where the elapsed time period t exceeds the first threshold t**1** and further increases, the controller **30** reduces the motor rotational speed R from the reduced rotational speed Rp to the idling rotational speed Ri in accordance with the elapsed time period t in a stepwise or continuous manner. Specifically, when the elapsed time period t reaches a second threshold t**2** (t=t**2**), the controller **30** controls the motor rotational speed R to the idling rotational speed Ri by further decreasing the motor rotational speed R. The controller **30** controls the motor rotational speed R to the idling rotational speed Ri by transmitting an instruction signal to the inverter **45** based on the calculated actual motor rotational speed Ra and the idling rotational speed Ri. The second threshold t**2** is a period longer than the first threshold t**1**.

[0085] Note that after the elapsed time period t reaches the first threshold t**1** and the motor rotational speed R is reduced to the reduced rotational speed Rp, the controller **30** maintains the motor rotational speed R at the reduced rotational speed Rp until the elapsed time period t reaches the second threshold t**2**. Then, the controller **30** decreases the motor rotational speed R to the idling rotational speed Ri in a stepwise or continuous manner after the elapsed time period t reaches the second threshold t**2**.

[0086] When it is determined that the manual operator **5** has been operated in a case where the elapsed time period t is equal to or greater than the first threshold t**1**, the controller **30** controls the motor rotational speed R to the set rotational speed Rt set immediately before the elapsed time period t reaches the first threshold t**1**.

[0087] The first threshold t**1** and the second threshold t**2** are, for example, predetermined values (periods) stored in advance in the storing device **31**. For example, the first threshold t**1** is defined to two seconds, and the second threshold t**2** is defined to four seconds.

[0088] Note that these values of the first threshold t**1** and the second threshold t**2** are merely illustrative and are not restrictive, and for example, the first threshold t**1** may be defined to three seconds and the second threshold t**2** may be defined to six seconds. The first threshold t**1** and/or the second threshold t**2** may be changeable to any values by operating an input device (e.g., a display, not illustrated) for inputting information that is connected to the controller **30**. In a case where priority is given to energy saving, the first threshold t**1** and the second threshold t**2** are desirably defined to short values, whereas in a case where priority is given to operability, the first threshold t**1** and the second threshold t**2** are desirably defined to long values. Furthermore, the first threshold t**1** and/or the second threshold t**2** may be changeable in accordance with the mode of the controller **30**. For example, in a case where the controller **30** is switchable among a plurality of modes (e.g., an eco-mode, a normal mode, and an efficiency mode) by a mode switching switch (not illustrated) included in the manual operator **5**, the first threshold t**1** and/or the second threshold t**2** are defined to increase in the order of the eco-mode, the normal mode, and the efficiency mode.

[0089] The predetermined value r is a constant rotational speed R defined in advance. The predetermined value r is, for example, a predetermined value (rotational speed) stored in advance in the storing device **31**. The predetermined value r is defined to be less than a difference between the lower limit value Rtmin (1000 rpm/min) of the set rotational speed Rt that can be set by the rotational speed operation actuator **36** and the idling rotational speed Ri (r<Rtmin−Ri). In the present example embodiment, the predetermined value r is defined to 500 rpm/min.

[0090] Note that this value of the predetermined value r is merely illustrative and is not restrictive as long as the value of the predetermined value r is defined to be less than a difference between the lower limit value Rtmin and the idling rotational speed Ri, and may be, for example, defined to 700 rpm/min. The predetermined value r may be changeable to any value by operating an input device (e.g., a display, not illustrated) for inputting information that is connected to the controller **30**. In a case where priority is given to energy saving, the predetermined value r is desirably defined to a large value, whereas in a case where priority is given to operability, the predetermined value r is desirably defined to a small value. The predetermined value r may be changeable in accordance with the mode of the controller **30**, as with the first threshold t**1** and/or the second threshold t**2**. For example, the predetermined value r is defined to decrease in the order of the eco-mode, the normal mode, and the efficiency mode. The predetermined value r may be calculated in accordance with the set rotational speed Rt set by the rotational speed operation actuator **36**. The controller **30** may, for example, calculate the predetermined value r based on a predetermined arithmetic expression stored in the storing device **31** so that the value of the predetermined value r increases as the set rotational speed Rt increases.

[0091] Next, a specific example of auto idling control performed by the controller **30** is described with reference to FIGS. **3** and **4**. FIGS. **3** and **4** illustrate a first example and a second example of the motor rotational speed R in a case where the auto idling control is performed, respectively. In the first example illustrated in FIG. **3**, a user sets the set rotational speed Rt to 2200 rpm/min by operating the rotational speed operation actuator **36**. Furthermore, in the first example illustrated in FIG. **3**, the user operates the manual operator **5** after the elapsed time period t exceeds the second threshold t**2** (at an elapsed time period t**3** illustrated in FIG. **3**, t**3**>t**2**).

[0092] During a first period T**1** illustrated in FIG. **3**, the user is operating the manual operator **5**, and the AI-SW **35***b* detects an ON state. Accordingly, the controller **30** determines that the manual operator **5** is being operated and any of the working devices **20** and **10** is in the actuated state. Since the elapsed time period t is zero and is less than the first threshold t**1** (t=0<t**1**), the controller **30** controls the motor rotational speed R to the set rotational speed Rt set by the rotational speed operation actuator **36**.

[0093] During a second period T**2**, a third period T**3**, and a fourth period T**4** illustrated in FIG. **3**, the user is not operating the manual operator **5**, and the AI-SW **35***b* detects an OFF state. In the first example illustrated in FIG. **3**, the user operates the manual operator **5** after the elapsed time period t exceeds the second threshold t**2**. Specifically, at the end of the fourth period T**4**, the user operates the manual operator **5**, and the AI-SW **35***b* detects an ON state.

[0094] In a case where the AI-SW **35***b* detects an OFF state, the controller **30** determines that the manual operator **5** is not being operated and all of the working devices **20** and **10** are in the non-actuated state, and calculates the elapsed time period t. The second period T**2** is a period where the elapsed time period t is longer than zero and is less than the first threshold **t1** ($0<t<t1$), and during the second period T**2**, the controller **30** controls the motor rotational speed R at the set rotational speed Rt (2200 rpm/min) set by the rotational speed operation actuator **36** continuously from the first period T**1**.

[0095] The third period T**3** is a period where the elapsed time period t is equal to or greater than the first threshold **t1** and is less than the second threshold **t2** ($t1 \le t < t2$), and during the third period T**3**, the controller **30** controls the motor rotational speed R to the reduced rotational speed Rp by reducing the motor rotational speed R by the predetermined value r (500 rpm/min) from the set rotational speed Rt set by the rotational speed operation actuator **36**. In the first example illustrated in FIG. **3**, the set rotational speed Rt is set to 2200 rpm/min, and therefore the controller **30** controls the motor rotational speed R to 1700 rpm/min.

[0096] The fourth period T**4** is a period from a time at which the elapsed time period t reaches the second threshold **t2** to a time at which the user operates the manual operator **5** ($t2 \le t \le t3$). During the fourth period T**4**, the controller **30** reduces the motor rotational speed R to the idling rotational speed Ri (250 rpm/min).

[0097] At the end of the fourth period T**4** ($t=t3$), the AI-SW **35***b* detects an ON state. Accordingly, the controller **30** determines that the manual operator **5** is being operated and any of the working devices **20** and **10** is in the actuated state. Accordingly, the controller **30** resets the calculated elapsed time period t and controls the motor rotational speed R to the set rotational speed Rt (2200 rpm/min) set immediately before the elapsed time period t reaches the first threshold **t1** (fifth period T**5**). That is, at the time of shift from the fourth period T**4** to the fifth period T**5**, the controller **30** increases the rotational speed R by 1950 rpm/min from the idling rotational speed Ri (250 rpm/min), to which the motor rotational speed R is reduced during the fourth period T**4**, to the set rotational speed Rt (2200 rpm/min).

[0098] In the second example illustrated in FIG. **4**, the user sets the set rotational speed Rt to 1600 rpm/min by operating the rotational speed operation actuator **36**. Furthermore, in the second example illustrated in FIG. **4**, the user operates the manual operator **5** after the elapsed time period t exceeds the first threshold **t1** and before the elapsed time period t reaches the second threshold **t2** (at an elapsed time period **t4** illustrated in FIG. **4**, $t1<t4<t2$).

[0099] During a sixth period T**6** illustrated in FIG. **4**, the user is operating the manual operator **5**, and the elapsed time period t is zero and is less than the first threshold **t1** ($t=0<t1$), and therefore the controller **30** controls the motor rotational speed R to the set rotational speed Rt set by the rotational speed operation actuator **36**.

[0100] During a seventh period T**7** and an eighth period T**8** illustrated in FIG. **4**, the user is not operating the manual operator **5**, and the AI-SW **35***b* detects an OFF state. The seventh period T**7** is a period where the elapsed time period t is longer than zero and is less than the first threshold **t1** ($0<t<t1$), and during the seventh period T**7**, the controller **30** controls the motor rotational speed R to the set rotational speed Rt (1600 rpm/min) set by the rotational speed operation actuator **36** continuously from the sixth period T**6**. The eighth period T**8** is a period where the elapsed time period t is equal to or greater than the first threshold **t1**, and ends when the user operates the manual operator **5** ($t1 \le t \le t4$). During the eighth period T**8**, the controller **30** controls the motor rotational speed R to the reduced rotational speed Rp by reducing the motor rotational speed R by the predetermined value r (500 rpm/min) from the set rotational speed Rt set by the rotational speed operation actuator **36**. In the second example illustrated in FIG. **4**, the set rotational speed Rt is set to 1600 rpm/min, and therefore the controller **30** controls the motor rotational speed R to 1100 rpm/min.

[0101] At the end of the eighth period T**8** ($t=t4$), the AI-SW **35***b* detects an ON state, and the controller **30** resets the calculated elapsed time period t and controls the motor rotational speed R to the set rotational speed Rt set immediately before the elapsed time period t reaches the first threshold **t1**. That is, at the time of shift from the eighth period T**8**, the controller **30** increases the rotational speed R by the predetermined value r (500 rpm/min) from 1100 rpm/min, which is the motor rotational speed R decreased during the eighth period T**8**, to the set rotational speed Rt.

[0102] As described above, in a case where the working devices **20** and **10** are in the non-actuated state and the elapsed time period t is equal to or greater than the second threshold **t2**, the controller **30** decreases the motor rotational speed R to the idling rotational speed Ri in two steps when the elapsed time period t reaches the first threshold **t1** and when the elapsed time period t reaches the second threshold **t2**. That is, the controller **30** decreases the motor rotational speed R to the idling rotational speed Ri in two stages in accordance with the elapsed time period t.

[0103] Accordingly, in a case where the user does not operate the manual operator **5** and the elapsed time period t reaches the second threshold **t2**, the controller **30** decreases the motor rotational speed R to the idling rotational speed Ri, and therefore energy saving of the working machine **1** can be achieved. Not only energy saving can be achieved by reducing the motor rotational speed R once before the elapsed time period t reaches the second threshold **t2** in a case where the elapsed time period t exceeds the first threshold **t1** and further increases, but also both response and energy saving of the working machine **1** can be achieved since original work can be resumed by increasing the motor rotational speed R by the predetermined value r by the controller **30** when the manual operator **5** is operated before the elapsed time period t reaches the second threshold **t2**.

[0104] Note that although control in which the controller **30** decreases the motor rotational speed R to the idling rotational speed Ri in two stages in accordance with the elapsed time period t has been described as an example in the present example embodiment, the controller **30** need just decrease the motor rotational speed R when the elapsed time period t reaches the first threshold **t1** and further decrease the motor rotational speed R to the idling rotational speed Ri in accordance with the elapsed time period t.

[0105] That is, the controller **30** may decrease the motor rotational speed R to the idling rotational speed Ri in three or four stages in accordance with the elapsed time period t instead of decreasing the motor rotational speed R to the idling rotational speed Ri in two stages in accordance with the elapsed time period t, and the number of stages in which the motor rotational speed R is reduced to the idling rotational speed Ri is not limited to two.

[0106] The controller **30** may control the motor rotational speed R to the reduced rotational speed Rp when the elapsed time period t reaches the first threshold **t1**, and decrease the motor rotational speed R successively from the reduced rotational speed Rp in accordance with the elapsed time period t such that the motor rotational speed R reaches the idling rotational speed Ri when the elapsed time period t reaches the second threshold t**2**.

[0107] Instead of the above idling control, the controller **30** may perform, depending on the mode, control of decreasing the motor rotational speed R from the set rotational speed Rt to the idling rotational speed Ri when the elapsed time period t reaches a predetermined threshold instead of decreasing the motor rotational speed R in a plurality of stages.

[0108] As illustrated in FIG. **1**, the working machine **1** includes the unloading lever **37**. The unloading lever **37** is a manual operator **5** that can be switched between a loading position (first position) where the working device **20** is permitted to work and an unloading position (second position) where the working device **20** is not permitted to work (prohibited from working). The unloading lever **37** is, for example, provided beside the operator's seat **4** so as to be swingable up and down. The unloading lever **37** closes a passage through which the user gets on to take the operator's seat **4** and gets out of the operator's seat **4** when the unloading lever **37** is swung down and is positioned in the loading position (first position, lowered position). On the other hand, the passage is opened when the unloading lever **37** is swung up and is positioned in the unloading position (second position, raised position).

[0109] When the unloading lever **37** is swung up and the unloading lever **37** is positioned in the second position during drive of the electric motor **46**, the controller **30** controls the electric motor **46** to a stop rotational speed R**0** (e.g., an extremely low rotational speed R such as 0 rpm/min or less than 1 rpm/min) corresponding to a stop state of the electric motor **46** irrespective of an operation of the rotational speed operation actuator **36** and idling control. The controller **30** decreases the rotational speed R of the electric motor **46** to the stop rotational speed R**0** by transmitting an instruction signal to the inverter **45** based on the calculated actual motor rotational speed Ra.

[0110] Note that in a case where the unloading lever **37** is swung up and the unloading lever **37** is positioned in the second position during drive of the electric motor **46**, the controller **30** may control the electric motor **46** to the idling rotational speed Ri. In this case, the controller **30** may control the electric motor **46** from the idling rotational speed Ri to the stop rotational speed R**0** in a case where a predetermined period elapses from the upward swinging operation of the unloading lever **37**.

[0111] Furthermore, in a case where the unloading lever **37** is swung up and the unloading lever **37** is positioned in the second position, the controller **30** may first control the electric motor **46** so that the motor rotational speed R decreases from the set rotational speed Rt set by the rotational speed operation actuator **36** by a predetermined rotational speed r' (r'>r) higher than the predetermined value r. The predetermined rotational speed r' is defined to be less than a difference between the lower limit value Rtmin of the set rotational speed Rt and the idling rotational speed Ri (r'<Rtmin-Ri). In this case, the controller **30** may use processing of controlling the motor rotational speed R to the idling rotational speed Ri when the elapsed time period t reaches the first threshold t**1** (t=t**1**) and controlling the motor rotational speed R to the stop rotational speed R**0** when the elapsed time period t reaches the second threshold t**2** (t=t**2**).

[0112] A flow of a series of processes of control of the motor rotational speed R performed by the controller **30** is described below with reference to FIG. **5**. FIG. **5** is a view for explaining a flow of a series of processes of control of the motor rotational speed R performed by the controller **30**. The series of processes illustrated in FIG. **5** are executed by the CPU based on a software program stored in advance in the memory of the controller **30**. FIG. **5** illustrates a state where the electric motor **46** is being driven, i.e., a state where the starter switch **32** is ON, and the controller **30** starts the electric motor **46** by the inverter **45** when the starter switch **32** is turned ON by the user. During drive of the electric motor **46**, the controller **30** determines whether or not the unloading lever **37** is in the first position (S**1**). In a case where the controller **30** determines that the unloading lever **37** is in the first position (S**1**, Yes), the controller **30** acquires an operation signal of the rotational speed operation actuator **36** and controls the rotational speed R of the electric motor **46** to the set rotational speed Rt (S**2**). Furthermore, the controller **30** determines whether or not the working devices **20** and **10** are working (S**3**).

[0113] In a case where the AI-SW **35***b* detects an OFF state and the controller **30** determines that all of the working devices **20** and **10** are in the non-actuated state (S**3**, No), the controller **30** determines that the elapsed time period t stored in the memory is zero (S**4**). In a case where the controller **30** determines that the elapsed time period t stored in the memory is zero (S**4**, Yes), the controller **30** starts calculation of the elapsed time period t (S**5**) and proceeds to S**1**.

[0114] In a case where the controller **30** determines that the elapsed time period t stored in the memory is not zero (S**4**, No), the controller **30** determines whether or not the elapsed time period t is equal to or greater than the first threshold t**1** (S**6**). In a case where the controller **30** determines that the elapsed time period t is less than the first threshold t**1** (S**6**, No), the controller **30** proceeds to S**1**.

[0115] On the other hand, in a case where the controller **30** determines that the elapsed time period t is equal to or greater than the first threshold t**1** (S**6**, Yes), the controller **30** determines whether or not the elapsed time period t is equal to or greater than the second threshold t**2** (S**7**). In a case where the controller **30** determines that the elapsed time period t is less than the second threshold t**2** (S**7**, No), the controller **30** transmits an instruction signal to the inverter **45** and thus decreases the motor rotational speed R to the reduced rotational speed Rp by subtracting the predetermined value r from the set rotational speed Rt (S**8**, first step), and proceeds to S**1**.

[0116] In a case where the controller **30** determines that the elapsed time period t is equal to or greater than the second threshold t**2** (S**7**, Yes), the controller **30** transmits an instruction signal to the inverter **45** and thus decreases the motor rotational speed R from the reduced rotational speed Rp to the idling rotational speed Ri (S**9**, second step), and proceeds to S**1**.

[0117] Note that in a case where the controller **30** determines in S**1** that the unloading lever **37** is not in the first position but in the second position (S**1**, No), the controller **30** resets the calculated elapsed time period t (S**10**), transmits an instruction signal to the inverter **45** and thus controls the motor rotational speed R to the stop rotational speed R**0** (S**11**), and proceeds to S**1**.

[0118] In a case where the AI-SW **35***b* detects an ON state and the controller **30** determines that any of the working devices **20** and **10** are in the actuated state in S**3** (S**3**, Yes), the controller **30** resets the calculated elapsed time period t (S**12**), and proceeds to S**2**.

[0119] Note that an example in which the electric motor **46** is driven by using electric power supplied from the battery unit **40** has been described in the present example embodiment, this is not restrictive, and the electric motor **46** may be driven, for example, by using electric power supplied from an external power source.

[0120] Although an example of the electric working machine **1** including the electric motor **46** as a rotary drive source has been described in the present example embodiment, the rotary drive source is not limited to the electric motor **46**. For example, the present example embodiment may be applied to a working machine including an engine (rotary drive source) such as a diesel engine or a gasoline engine instead of the electric motor **46**. The present example embodiment may be also applied to a hybrid-type working machine using a combination of an electric motor (rotary drive source) and an engine (rotary drive source).

[0121] Although a configuration in which when the elapsed time period t reaches the first threshold t**1**, the motor rotational speed R is reduced by the predetermined value r from the target rotational speed (set rotational speed) Rt set by the rotational speed operation actuator **36** has been described in the present example embodiment, this is not restrictive. For example, when the elapsed time period t reaches the first threshold t**1**, the motor rotational speed R may be reduced by the predetermined value r from an actual motor rotational speed (actual rotational speed) Ra at a time immediately before the elapsed time period t detected by the rotational speed detector **35***a* reaches the first threshold t**1**. In a case where the rotary drive source is an engine, an actual rotational speed Ra of the engine may be detected by the rotational speed detector **35***a*, and the motor rotational speed R may be decreased by a predetermined value from the actual rotational speed Ra at a time immediately before the elapsed time period t reaches the first threshold t**1**.

[0122] Example embodiments of the present invention provide working machines **1** and methods of controlling working machines **1** described in the following items.

[0123] (Item 1) A working machine **1** including a machine body **2**, a rotary drive source **46**, a hydraulic device P**1** to be actuated by power generated by the rotary drive source **46**, a working device **20**, **10** to be actuated by a hydraulic pressure of hydraulic fluid supplied from the hydraulic device P**1**, a work detector **35***b* to detect whether or not the working device **20**, **10** is actuated, and a controller **30** configured or programmed to, when an elapsed time period t which starts when the working device **20**, **10** enters a non-actuated state reaches a first threshold t**1**, reduce a rotational speed R of the rotary drive source **46** to a reduced rotational speed Rp obtained by subtracting a predetermined value r from a set rotational speed Rt at a time immediately before the elapsed time period t reaches the first threshold t**1** or from an actual rotational speed R, and, as the elapsed time period t exceeds the first threshold t**1** and further increases, reduce the rotational speed R of the rotary drive source **46** from the reduced rotational speed Rp to an idling rotational speed Ri in accordance with the elapsed time period in a stepwise or continuous manner.

[0124] With the working machine **1** according to item 1, the rotational speed R of the rotary drive source **46** is first reduced to the reduced rotational speed Rp and then further reduced to the idling rotational speed Ri, instead of reducing the rotational speed R of the rotary drive source **46** to the idling rotational speed Ri at one time. Accordingly, in a case where the elapsed time period t exceeds the first threshold t**1** and further increases and the rotational speed R of the rotary drive source **46** is reduced to the idling rotational speed Ri, the energy source consumed by driving the rotary drive source **46** can be reduced, whereas in a case where the elapsed time period t is relatively short and the rotational speed R of the rotary drive source **46** has not decreased to the idling rotational speed Ri, the rotational speed R of the rotary drive source **46** can be easily increased.

[0125] (Item 2) The working machine **1** according to item 1, further including a manual operator **5** to be operated to control the working device **20**, **10**, wherein the work detector **35***b* is operable to detect whether or not the working device **20**, **10** is actuated based on the operation of the manual operator **5**.

[0126] With the working machine **1** according to item 2, the work detector **35***b* can more reliably detect whether or not the working device **20** is actuated.

[0127] (Item 3) The working machine **1** according to item 1 or 2, further including a rotational speed operation actuator **36** to be operated to set the set rotational speed Rt, wherein the controller **30** is configured or programmed to control the rotational speed R of the rotary drive source **46** at the set rotational speed Rt set by operating the rotational speed operation actuator **36** in a case that the elapsed time period t is less than the first threshold t**1**, and reduce the rotational speed R of the rotary drive source **46** from the set rotational speed Rt to the reduced rotational speed Rp when the elapsed time period t reaches the first threshold t**1**.

[0128] With the working machine **1** according to item 3, when the elapsed time period t reaches the first threshold t**1**, the rotational speed R of the rotary drive source **46** is reduced from the rotational speed Rt set by operating the rotational speed operation actuator **36**, and therefore the energy source consumed by driving the rotary drive source **46** can be reduced without having to operate the rotational speed operation actuator **36**.

[0129] (Item 4) The working machine **1** according to item 1 or 2, further including a rotational speed detector **35***a* to detect the rotational speed R of the rotary drive source **46**, wherein the controller **30** is configured or programmed to, when the elapsed time period t reaches the first threshold t**1**, reduce the rotational speed R of the rotary drive source **46** to the reduced rotational speed Rp from a rotational speed R of the rotary drive source **46** detected by the rotational speed detector **35***a* immediately before the elapsed time period t reaches the first threshold t**1**.

[0130] With the working machine **1** according to item 4, when the elapsed time period t reaches the first threshold t**1**, the rotational speed R of the rotary drive source **46** is reduced from the rotational speed R of the rotary drive source **46** detected immediately before the elapsed time period t reaches the first threshold t**1**, and therefore the energy source consumed by driving the rotary drive source **46** can be reduced without having to operate the rotational speed operation actuator **36**.

[0131] (Item 5) The working machine **1** according to any one of items 1 to 4, wherein the controller **30** is configured or programmed to, after the elapsed time period t reaches the first threshold t**1** and the rotational speed R of the rotary drive source **46** is reduced to the reduced rotational speed

Rp, maintain the rotational speed R of the rotary drive source **46** at the reduced rotational speed Rp until the elapsed time period t reaches a second threshold t**2** greater than the first threshold t**1**, and when the elapsed time period t reaches the second threshold t**2**, further reduce the rotational speed R of the rotary drive source **46** from the reduced rotational speed Rp.

[0132] With the working machine **1** according to item 5, the rotational speed R of the rotary drive source **46** can be easily increased before the elapsed time period t reaches the second threshold t**2**, and after the elapsed time period t reaches the second threshold t**2**, the energy source consumed by driving the rotary drive source **46** can be further reduced.

[0133] (Item 6) The working machine **1** according to item 5, wherein the controller **30** is configured or programmed to reduce the rotational speed R of the rotary drive source **46** to the idling rotational speed Ri in a stepwise or continuous manner after the elapsed time period t reaches the second threshold t**2**.

[0134] With the working machine **1** according to item 6, the energy source consumed by driving the rotary drive source **46** can be reduced more reliably after the elapsed time period t reaches the second threshold t**2**.

[0135] (Item 7) The working machine **1** according to any one of items 1 to 6, wherein the idling rotational speed Ri is a rotational speed R of the rotary drive source **46** for a case where the working device **20**, **10** does not perform work, and the rotary drive source **46** driven at the idling rotational speed Ri does not generate enough power to cause the hydraulic device P**1** to actuate the working device **20**, **10**.

[0136] With the working machine **1** according to item 7, when the elapsed time period t reaches the second threshold t**2**, the energy source consumed by driving the rotary drive source **46** can be reduced more reliably, and also it is possible to eliminate or reduce the likelihood that the working device(s) **20**, **10** will operate unintentionally.

[0137] (Item 8) The working machine **1** according to item 2 or according to any one of claims **3** to **7** taken in combination with item 2, wherein the controller **30** is configured or programmed to, when the controller **30** determines that the manual operator **5** is operated when the elapsed time period t is equal to or greater than the first threshold t**1**, control the rotational speed R of the rotary drive source **46** to the set rotational speed Rt at a time immediately before the elapsed time period t reaches the first threshold t**1**.

[0138] With the working machine **1** according to item 8, the rotational speed R of the rotary drive source **46** can be easily returned to the original rotational speed R.

[0139] (Item 9) The working machine **1** according to any one of items 1 to 8, wherein the rotary drive source **46** is an electric motor to be driven by electric power or an engine to be driven by burning fuel.

[0140] With the working machine **1** according to item 9, the above configuration makes it possible to reduce energy source (electric power) consumed by driving the electric motor and reduce energy source (fuel) consumed by driving the engine.

[0141] (Item 10) A method of controlling a working machine **1** including a machine body **2**, a rotary drive source **46**, a hydraulic device P**1** to be actuated by power generated by the rotary drive source **46**, and a working device **20**, **10** to be actuated by a hydraulic pressure of hydraulic fluid supplied from the hydraulic device P**1**, the method including a first step including, when an elapsed time period t which starts when the working device **20**, **10** enters a non-actuated state reaches a first threshold t**1**, reducing a rotational speed R of the rotary drive source **46** to a reduced rotational speed Rp obtained by subtracting a predetermined value r from a set rotational speed Rt at a time immediately before the elapsed time period t reaches the first threshold t**1** or from an actual rotational speed R, and a second step including, as the elapsed time period t exceeds the first threshold t**1** and further increases, reducing the rotational speed R of the rotary drive source **46** from the reduced rotational speed Rp to an idling rotational speed Ri in accordance with the elapsed time period t in a stepwise or continuous manner.

[0142] With the method for controlling a working machine **1** according to item 10, the rotational speed R of the rotary drive source **46** is first reduced to the reduced rotational speed Rp and then further reduced to the idling rotational speed Ri, instead of reducing the rotational speed R of the rotary drive source **46** to the idling rotational speed Ri at one time. Accordingly, in a case where the elapsed time period t exceeds the first threshold t**1** and further increases and the rotational speed R of the rotary drive source **46** is reduced to the idling rotational speed Ri, the energy source consumed by driving the rotary drive source **46** can be reduced, whereas in a case where the elapsed time period t is relatively short and the rotational speed R of the rotary drive source **46** has not decreased to the idling rotational speed Ri, the rotational speed R of the rotary drive source **46** can be easily increased.

[0143] While example embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

What is claimed is:

1. A working machine comprising:

a machine body;

a rotary drive source;

a hydraulic device to be actuated by power generated by the rotary drive source;

a working device to be actuated by a hydraulic pressure of hydraulic fluid supplied from the hydraulic device;

a work detector to detect whether or not the working device is actuated; and

a controller configured or programmed to, when an elapsed time period which starts when the working device enters a non-actuated state reaches a first threshold, reduce a rotational speed of the rotary drive source to a reduced rotational speed obtained by subtracting a predetermined value from a set rotational speed at a time immediately before the elapsed time period reaches the first threshold or from an actual rotational speed, and, as the elapsed time period exceeds the first threshold and further increases, reduce the rotational speed of the rotary drive source from the reduced rotational speed to an idling rotational speed in accordance with the elapsed time period in a stepwise or continuous manner.

2. The working machine according to claim **1**, further comprising a manual operator to be operated to control the working device; wherein the work detector is operable to detect whether or not the working device is actuated based on the operation of the manual operator.

3. The working machine according to claim **2**, further comprising a rotational speed operation actuator to be operated to set the set rotational speed; wherein

the controller is configured or programmed to control the rotational speed of the rotary drive source at the set rotational speed set by operating the rotational speed operation actuator in a case that the elapsed time period is less than the first threshold, and reduce the rotational speed of the rotary drive source from the set rotational speed to the reduced rotational speed when the elapsed time period reaches the first threshold.

4. The working machine according to claim **2**, further comprising a rotational speed detector to detect the rotational speed of the rotary drive source; wherein

the controller is configured or programmed to, when the elapsed time period reaches the first threshold, reduce the rotational speed of the rotary drive source to the reduced rotational speed from a rotational speed of the rotary drive source detected by the rotational speed detector immediately before the elapsed time period reaches the first threshold.

5. The working machine according to claim **1**, wherein

the controller is configured or programmed to:

after the elapsed time period reaches the first threshold and the rotational speed of the rotary drive source is reduced to the reduced rotational speed, maintain the rotational speed of the rotary drive source at the reduced rotational speed until the elapsed time period reaches a second threshold greater than the first threshold; and

when the elapsed time period reaches the second threshold, further reduce the rotational speed of the rotary drive source from the reduced rotational speed.

6. The working machine according to claim **5**, wherein

the controller is configured or programmed to reduce the rotational speed of the rotary drive source to the idling rotational speed in a stepwise or continuous manner after the elapsed time period reaches the second threshold.

7. The working machine according to claim **6**, wherein

the idling rotational speed is a rotational speed of the rotary drive source for a case where the working device does not perform work, and the rotary drive source driven at the idling rotational speed does not generate enough power to cause the hydraulic device to actuate the working device.

8. The working machine according to claim **2**, wherein

the controller is configured or programmed to, when the controller determines that the manual operator is operated when the elapsed time period is equal to or greater than the first threshold, control the rotational speed of the rotary drive source to the set rotational speed at a time immediately before the elapsed time period reaches the first threshold.

9. The working machine according to claim **1**, wherein

the rotary drive source is an electric motor to be driven by electric power or an engine to be driven by burning fuel.

10. A method of controlling a working machine including a machine body, a rotary drive source, a hydraulic device to be actuated by power generated by the rotary drive source, and a working device to be actuated by a hydraulic pressure of hydraulic fluid supplied from the hydraulic device, the method comprising:

a first step including, when an elapsed time period which starts when the working device enters a non-actuated state reaches a first threshold, reducing a rotational speed of the rotary drive source to a reduced rotational speed obtained by subtracting a predetermined value from a set rotational speed at a time immediately before the elapsed time period reaches the first threshold or from an actual rotational speed; and

a second step including, as the elapsed time period exceeds the first threshold and further increases, reducing the rotational speed of the rotary drive source from the reduced rotational speed to an idling rotational speed in accordance with the elapsed time period in a stepwise or continuous manner.

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